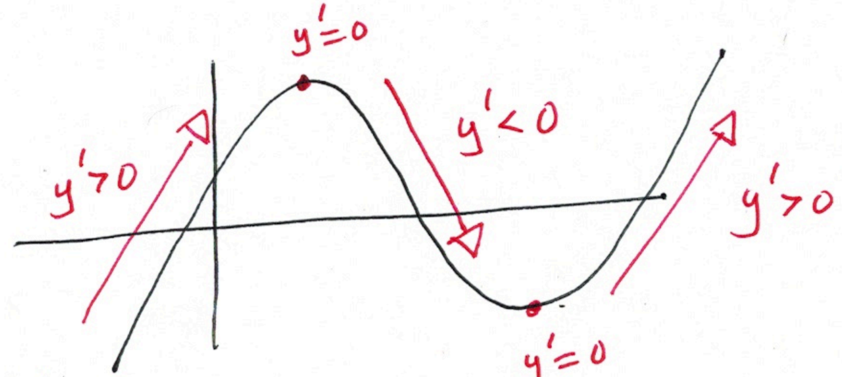
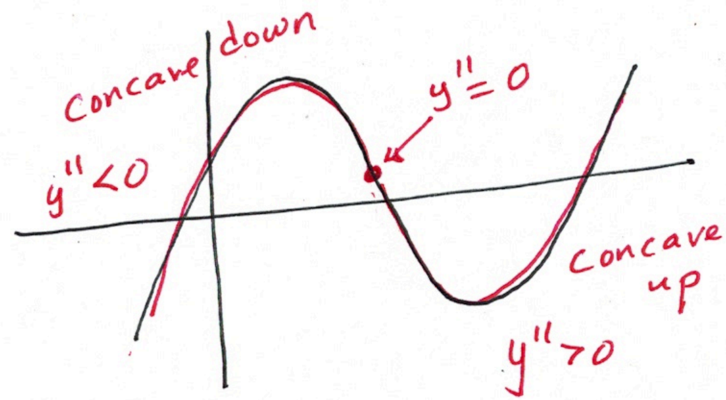
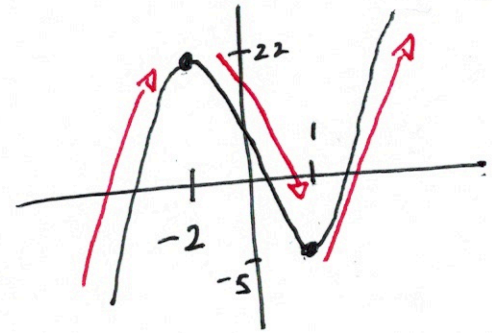
Section 4.3 Derivatives And Graphing





Ex ① $f(x) = 2x^3 + 3x^2 - 12x + 2$



$f'(x) = 6x^2 + 6x - 12$

$0 = 6x^2 + 6x - 12$

$0 = 6(x^2 + x - 2)$

$0 = 6(x-1)(x+2)$

$x = 1 \qquad x = -2$

Increasing $(-\infty, -2) \cup (1, \infty)$

Decreasing $(-2, 1)$

Ex ① cont'd

$$f'(x) = 6x^2 + 6x - 12$$

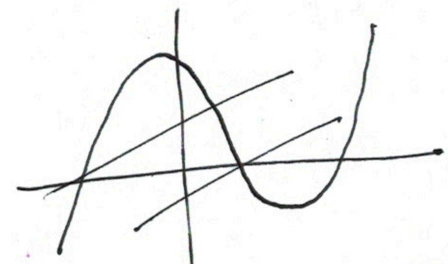
$$f''(x) = 12x + 6$$

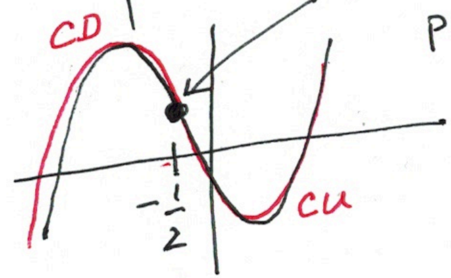
$$0 = 12x + 6$$

$$-6 = 12x$$

$$\frac{-6}{12} = x$$

$$-\frac{1}{2} = x$$





inflection point $(-.5, 8.5)$

Concave Down $(-\infty, -\frac{1}{2})$

Concave Up $(-\frac{1}{2}, \infty)$

Ex ② $f(x) = x + 3\sin(2x)$ on $[0, 2\pi]$

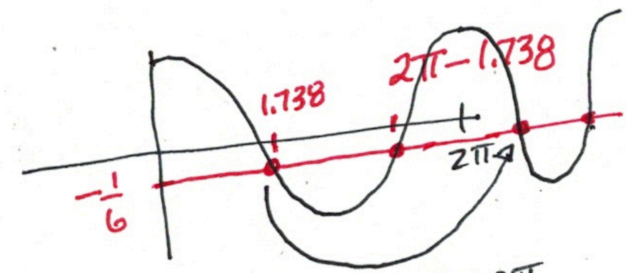
where is $f(x)$ increasing?

$$f'(x) = 1 + 3\cos(2x)\cdot 2$$

$$0 = 1 + 6\cos(2x)$$

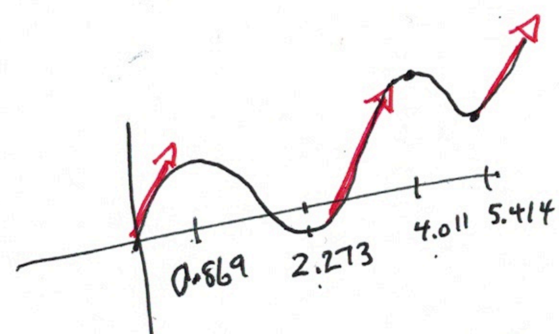
$$-1 = 6\cos(2x)$$

$$-\frac{1}{6} = \cos(2x)$$



$2x = 1.738$ $\qquad$ $2x = 2\pi - 1.738$

$x = 0.869$ $\qquad$ $x = 2.273$



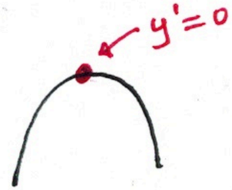
$2x = 1.738 + 2\pi$ $\qquad$ $2x = 4\pi - 1.738$

$x = 4.011$ $\qquad$ $x = 5.414$

Inc $[0, 0.869) \cup (2.273, 4.011) \cup (5.414, 2\pi]$

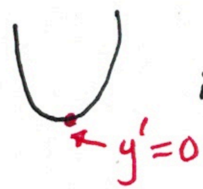
Second Derivative Test

Concave Down

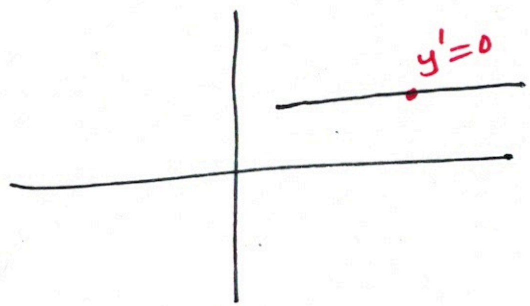


is at a maximum point

Concave up

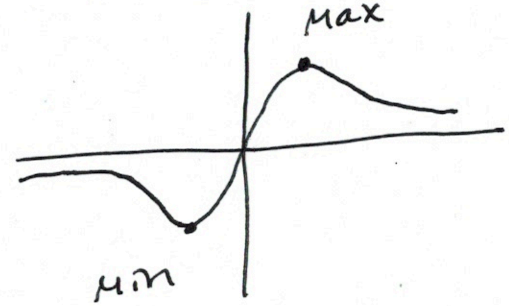


is at a minimum point.



Ex ③ Use 2^{nd} Derivative test to find extrema

$$f(x) = \frac{x}{x^2+1}$$



$$f'(x) = \frac{1\cdot(x^2+1) - x(2x)}{(x^2+1)^2}$$

$$f'(x) = \frac{-x^2+1 = 0}{(x^2+1)^2 = 0}$$

$f'(x)=0$

$-x^2+1=0$

$1 = x^2$

$\pm 1 = x$

$f'(x)$ DNE

$(x^2+1)^2 = 0$

$x^2+1=0$

$x^2 = -1$

$x = \pm\sqrt{-1}$

No solution

Ex ③ cont'd

$$f'(x) = \frac{-x^2+1}{(x^2+1)^2}$$

$$f''(x) = \frac{-2x(x^2+1)^2 - (-x^2+1)\cdot 2(x^2+1)(2x)}{(x^2+1)^4}$$

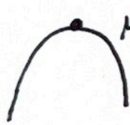
$$f''(x) = \frac{2x\cancel{(x^2+1)}\left[-(x^2+1) - 2(-x^2+1)\right]}{(x^2+1)^{\cancel{4}\,3}}$$

$$f''(x) = \frac{2x\left[x^2-3\right]}{(x^2+1)^3}$$

$\underline{x=1}$

$$f''(1) = \frac{2(1-3)}{(1+1)^3}$$

$$= \frac{-4}{8} \text{ Neg.}$$



max

$\underline{x=-1}$

$$f''(-1) = \frac{-2(1-3)}{(1+1)^3}$$

$$f''(-1) = \frac{4}{8} \text{ pos.}$$

min

$$\boxed{\begin{array}{l} \text{MAX } (1, \tfrac{1}{2}) \\ \\ \text{min } (-1, -\tfrac{1}{2}) \end{array}}$$